

**BOUGAINVILLE REEF, Australia**WMO: **952880**Lat: **15.488S**Lon: **147.118E**Elev: **43**StdP: **14.67**Time Zone: **10.00 (E10)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 8             | 70.1       | 71.5 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 33.8                  | 74.7 | 32.2 | 74.9 | 20.2                  | 140  | 0.736 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 2             | 3.4                    | 86.3            | N/A  | 85.5 | N/A  | 85.0 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 10.5                 | 320  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 33.1          | 31.0 | 29.3 | DB                         | 69.8 | 89.7               | 1.4 | 2.9                                                | 68.8 | 91.7       | 68.0 | 93.4       | 67.3 | 95.0       | 66.3 | 97.1 | (4) |
| (5)               |      |      | WB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  | N/A  | (5) |

**Monthly Climatic Design Conditions**

|                                                                                  |         | Annual                           | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|---------|----------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |         | (d)                              | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg   | 78.8                             | 82.3  | 82.7  | 81.8  | 80.2  | 78.3  | 76.2  | 75.1  | 74.6  | 75.8  | 77.6  | 79.9  | 81.8  |
|                                                                                  | DBStd   | 3.34                             | 2.03  | 1.74  | 2.10  | 1.84  | 1.92  | 1.66  | 1.56  | 1.54  | 1.53  | 1.70  | 1.62  | 1.73  |
|                                                                                  | HDD50   | 0                                | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | HDD65   | 0                                | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDD50   | 10525                            | 1000  | 916   | 985   | 907   | 878   | 786   | 777   | 764   | 773   | 857   | 897   | 985   |
|                                                                                  | CDD65   | 5050                             | 535   | 496   | 520   | 457   | 413   | 336   | 312   | 299   | 323   | 392   | 447   | 520   |
|                                                                                  | CDH74   | 45086                            | 6418  | 6047  | 6065  | 4764  | 3493  | 1823  | 1128  | 805   | 1407  | 2811  | 4392  | 5933  |
|                                                                                  | CDH80   | 8816                             | 2035  | 2075  | 1749  | 731   | 115   | 3     | 3     | 1     | 3     | 69    | 473   | 1559  |
| Wind                                                                             | WSAvg   | 18.2                             | 14.6  | 14.0  | 16.8  | 21.2  | 21.5  | 21.9  | 22.3  | 19.9  | 18.7  | 17.2  | 15.7  | 14.2  |
| Precipitation                                                                    | PrecAvg | 68.4                             | 11.9  | 14.4  | 13.7  | 6.2   | 3.3   | 2.3   | 1.9   | 1.4   | 1.1   | 2.0   | 2.9   | 7.4   |
|                                                                                  | PrecMax | 110.7                            | 26.4  | 33.7  | 34.4  | 22.7  | 11.0  | 5.3   | 4.5   | 7.7   | 4.0   | 7.8   | 12.8  | 20.1  |
|                                                                                  | PrecMin | 29.6                             | 1.1   | 0.7   | 0.8   | 0.1   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 1.4   |
|                                                                                  | PrecStd | 19.6                             | 6.5   | 7.9   | 8.4   | 4.9   | 2.2   | 1.3   | 1.1   | 1.5   | 1.1   | 2.0   | 3.5   | 5.2   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%    | DB                               | 86.8  | 87.6  | 88.4  | 84.2  | 82.7  | 80.5  | 79.1  | 79.1  | 80.3  | 82.8  | 84.9  | 86.8  |
|                                                                                  |         | MCWB                             | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 2%      | DB                               | 85.5  | 86.3  | 85.8  | 83.5  | 81.9  | 79.8  | 78.3  | 78.1  | 79.3  | 81.8  | 83.6  | 85.4  |
|                                                                                  |         | MCWB                             | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 5%      | DB                               | 85.1  | 85.6  | 85.0  | 83.1  | 81.2  | 79.2  | 77.9  | 77.4  | 78.6  | 80.8  | 82.8  | 84.6  |
|                                                                                  |         | MCWB                             | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 10%     | DB                               | 84.6  | 85.0  | 84.4  | 82.6  | 80.6  | 78.7  | 77.3  | 76.5  | 77.8  | 80.0  | 82.0  | 84.1  |
|                                                                                  |         | MCWB                             | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 0.4%    | WB                               | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |         | MCDB                             | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 2%      | WB                               | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |         | MCDB                             | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB   | MDBR                             | 3.4   | 3.4   | 3.4   | 2.8   | 2.8   | 2.1   | 2.2   | 1.8   | 1.5   | 1.7   | 2.1   | 2.6   |
|                                                                                  |         | MCDBR                            | 3.7   | 4.2   | 3.8   | 3.7   | 3.9   | 2.4   | 3.2   | 2.2   | 2.0   | 2.4   | 3.0   | 3.5   |
|                                                                                  |         | MCWBR                            | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 5% WB   | MCDBR                            | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |         | MCWBR                            | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |         | Clear-Sky<br>Solar<br>Irradiance | taub  | 0.395 | 0.391 | 0.398 | 0.393 | 0.384 | 0.381 | 0.364 | 0.368 | 0.380 | 0.395 | 0.394 |
| taud                                                                             | 2.595   |                                  | 2.619 | 2.584 | 2.578 | 2.567 | 2.565 | 2.597 | 2.572 | 2.519 | 2.491 | 2.530 | 2.570 |       |
| Ebn at Noon                                                                      | 301     |                                  | 300   | 291   | 282   | 273   | 268   | 277   | 285   | 292   | 295   | 299   | 301   |       |
| Edh at Noon                                                                      | 33      |                                  | 32    | 32    | 31    | 29    | 29    | 28    | 30    | 34    | 36    | 35    | 34    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg  | 2032                             | 1958  | 1769  | 1680  | 1474  | 1323  | 1421  | 1705  | 2030  | 2219  | 2287  | 2191  |       |
|                                                                                  | RadStd  | 190                              | 181   | 127   | 140   | 125   | 102   | 100   | 109   | 72    | 111   | 161   | 229   |       |

**Historical Trends**

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page